How BloodLink works, and how to test all of it

A blood-donation platform built around registered **blood banks**. Three roles, one spine: a **member** asks for blood, the system scores and ranks compatible members by availability and distance, a hospital **officer** matches one or more of them until the request is fulfilled, and everyone locates the nearest bank and follows the shortest route there. This manual walks each role end to end, maps how the pieces connect, explains the enhancements, and gives you a role-by-role QA script.

WHAT'S NEW IN V6

Server-checked donations (donor's own blood group, compatibility, open request, unit limits, cumulative fulfilment) · walk-in donations that add to bank stock · strict registration validation and unique Member IDs · **My Profile** and **My Donations** for members · officer dashboard with bank stock · live home-page figures · and an interactive **Blood Bank Locator** with a draggable "you are here" pin, the shortest road route drawn in blue, and kilometres on the line.

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01 / The cast

Three roles, at a glance

Everyone signs in through one of two doors — the admin portal (`login.php`) or the shared user portal (`user-login.php`, role dropdown: Member or Hospital Officer). The session role gates every page; opening another role's page sends you back to your own dashboard.

SESSION ROLE: ADMIN

Administrator

Runs the bank network: manages blood banks, members, hospital officers and per-bank stock; posts emergency alerts; monitors the request queue; searches the donor pool; reads charts and reports.

SESSION ROLE: OFFICER

Hospital Officer

Belongs to one blood bank. Works the request queue and the Emergency Board, matches requests to the best-scored available members, **confirms each donation**, and records **walk-in donations** that add units to their bank's stock.

SESSION ROLE: MEMBER

Member

One account that both **donates** and **requests**. Registers with a blood group and address, sets availability, checks the 56-day rule, files blood requests, sees alerts, keeps a profile, reviews their donation history and routes to the nearest bank.

WHY A SINGLE "MEMBER" ACCOUNT

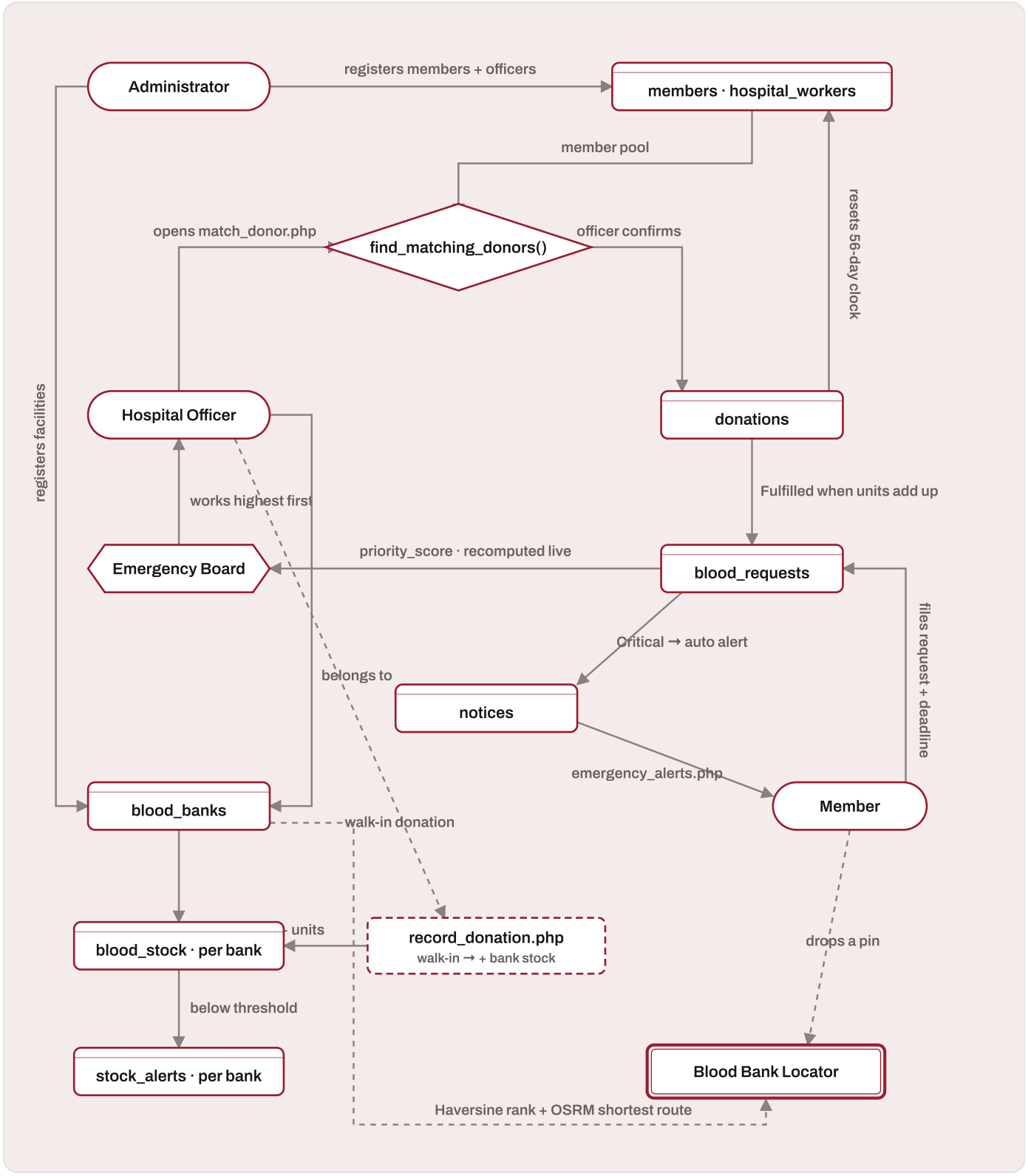
A real person can both give and need blood, so BloodLink uses **one member account for both** — every member is a searchable donor *and* can file requests. On top of that, four enhancements make it "enhanced": a **donor availability registry** with Find Donors, an **emergency workflow** that ranks requests by a live priority score, **compatibility-aware matching**, and **analytics dashboards** — plus the **Blood Bank Locator** (Section 06).

02 / The spine

How the whole app connects

Everything orbits one workflow: a request becomes one or more matched, recorded donations. The locator, the 56-day rule, availability and per-bank stock all hang off it.

- **Admins** register the `blood_banks`, the people, and each bank's `blood_stock`.
- **Members** register once and create `blood_requests` with an urgency and optional deadline. A Critical Emergency also writes a member-facing `notices` alert.
- **Officers** belong to a bank. The Emergency Board ranks open requests by a priority score recomputed on every load. `match_donor.php` runs `find_matching_donors()` (compatible, 56-day-eligible, requester excluded, available first, nearest). `process_donation()` writes the donation in one transaction, resets the donor's clock, and marks the request *Approved* or — once the units add up — *Fulfilled*.
- **Walk-in donations** (`record_donation.php`) go through the same function without a request; their units are added to the officer's bank stock and the bank's low-stock alert is re-checked.
- Everyone uses the **Blood Bank Locator**: drop a pin, banks rank nearest-first, and the shortest road route is drawn.


Administrator
registers members + officers
members · hospital_workers
member pool
resets 56-day clock
opens match_donor.php
find_matching_donors()
officer confirms
registers facilities
Hospital Officer
donations
works highest first
Fulfilled when units add up
Emergency Board
priority_score · recomputed live
blood_requests
Critical → auto alert
files request + deadline
belongs to
notices
emergency_alerts.php
Member
blood_banks
walk-in donation
units
record_donation.php
walk-in → + bank stock
drops a pin
blood_stock · per bank
below threshold
stock_alerts · per bank
Blood Bank Locator
Haversine rank + OSRM shortest route

03 / Data model

Nine tables, and who touches them

TABLE	HOLDS	WRITTEN BY	READ BY
`login`	Admin accounts	seed	admin login
`blood_banks`	Facilities (name — unique, address, geocoded lat/lng, contact)	admin (Blood Banks)	locator, stock, officers, reports
`hospital_workers`	Officers (+ required `blood_bank_id`, position; unique email)	admin (add officer)	officer login, donations, reports
`members`	Unified donor + recipient accounts: blood group, 56-day clock, lat/lng, availability, **unique** `member_code`, unique email	registration, admin, My Profile	login, matching, Find Donors, requests, reports
`blood_requests`	Requests: urgency, severity, `required_within_hours`, live `priority_score`, status, lat/lng; `recipient_id` → the member who asked	member (Request Blood)	queue, Emergency Board, match, history
`donations`	Donation ledger — **donor's** blood group, units, bank; `request_id` NULL for walk-ins	`process_donation()`	My Donations, histories, reports
`blood_stock`	Units per bank per blood type	admin, walk-in donations	stock page, dashboards, locator, reports
`stock_alerts`	Low-stock alerts per bank per type	stock page, walk-in donations	dashboards, stock page
`notices`	Emergency alerts, stored as plain text	officer / admin, auto (critical request)	member Emergency Alerts

This matches the brief's ERD (Figure 5.3): the design centres on `blood_banks`, with stock and alerts scoped per bank and officers belonging to a bank. Members with donations on record — and officers who confirmed donations — can't be deleted, so the ledger is never broken; mark a member unavailable instead.

04 / Role walkthroughs

What each role does, end to end

ADMINISTRATOR runs the network

Signs in at `login.php` (`admin` / `admin`) → `dashboard.php` .

- **Blood Banks:** add / edit / delete facilities. Each address is geocoded onto the map; a new bank gets an empty stock row per blood type. Duplicate names are refused with a message.
- **Members:** add, edit (including an optional password reset and the "Available to donate" switch) and delete. The same validation as registration applies.
- **Hospital Officers:** add officers — a blood bank is required, since donations are banked there. Duplicate emails are refused.
- **Blood Stock:** pick a bank, set units and the low-stock threshold per type (whole numbers, 0 or more). Dropping below threshold raises that bank's alert; restocking resolves it.
- **Oversight:** Emergency Board, Find Donors, Blood Requests, Emergency Alerts (post / delete any), the locator, and Reports with charts and a clean CSV export.

`HOSPITAL OFFICER` makes the match

Signs in at `user-login.php` (Hospital Officer, email + password) → `officer_dashboard.php`, which shows open and critical requests, the bank's low-stock count, units recorded this month, and the bank's stock by blood group.

- **Emergency Board:** open requests ranked by live priority, each with a countdown to its deadline.
- **Match & confirm:** `match_donor.php` lists compatible, eligible members (the requester excluded), available first, with a match score and distance. Unavailable members show a disabled button. Each confirmation is 1–2 units and never more than the request still needs; several donors can fulfil one request together.
- **Record Donation:** log a walk-in by the member's email or Member ID. The 56-day rule is checked and the units go into this officer's bank stock.
- **Emergency Alerts** (post / delete own), **Find Donors**, **Geo-Map**.

`MEMBER` is both supply and demand

Registers at `register.php`, signs in with email *or* Member ID + password → `member_dashboard.php`.

- **Registration rules:** name, valid email, 7–15 digit phone, one of the eight blood groups, password of 8+ characters with a letter and a number (typed twice), optional Member ID (letters, numbers, dashes; must be unique — blank auto-generates). Errors are listed and the form keeps what was typed.
- **My Availability:** status, schedule and contact preference; available members surface first.
- **Donate Blood / My Donations:** eligibility inside or outside the 56-day window, and every donation recorded for you — matched or walk-in — with totals.
- **Request Blood:** patient, group, units, hospital, urgency, optional deadline. Critical Emergency auto-posts an alert.
- **My Profile:** update name, email, phone and address (re-geocoded) and change your password. Blood group is changed by an officer or admin.
- **Request History**, **Emergency Alerts**, **Nearby Banks** (the locator).

05 / The enhancements

Four features that lift it above a basic register

01 · AVAILABILITY REGISTRY

Members say when they can give

Each member keeps an `is_available` flag, a schedule and a contact preference. **Find Donors** searches the pool by group and mode (compatible / exact), optionally available-only, ranked from a chosen bank.

02 · EMERGENCY WORKFLOW

Requests are triaged, not just queued

A request carries a severity and an optional deadline. The **Emergency Board** recomputes each `priority_score` from the time actually left, so a request climbs the board as its deadline approaches.

03 · COMPATIBILITY-AWARE MATCHING

The system scores every donor — and enforces it

A compatibility score rewards exact matches and universal O−, and available members sort first. On confirmation the server re-checks compatibility, eligibility and availability, and records the **donor's** blood group.

04 · ANALYTICS

At-a-glance charts

The admin dashboard charts stock by blood group; Reports adds a member blood-group distribution and a donation-status doughnut, KPIs and CSV export. The home page shows live member, unit and bank counts.

HOW THE PRIORITY SCORE IS COMPUTED

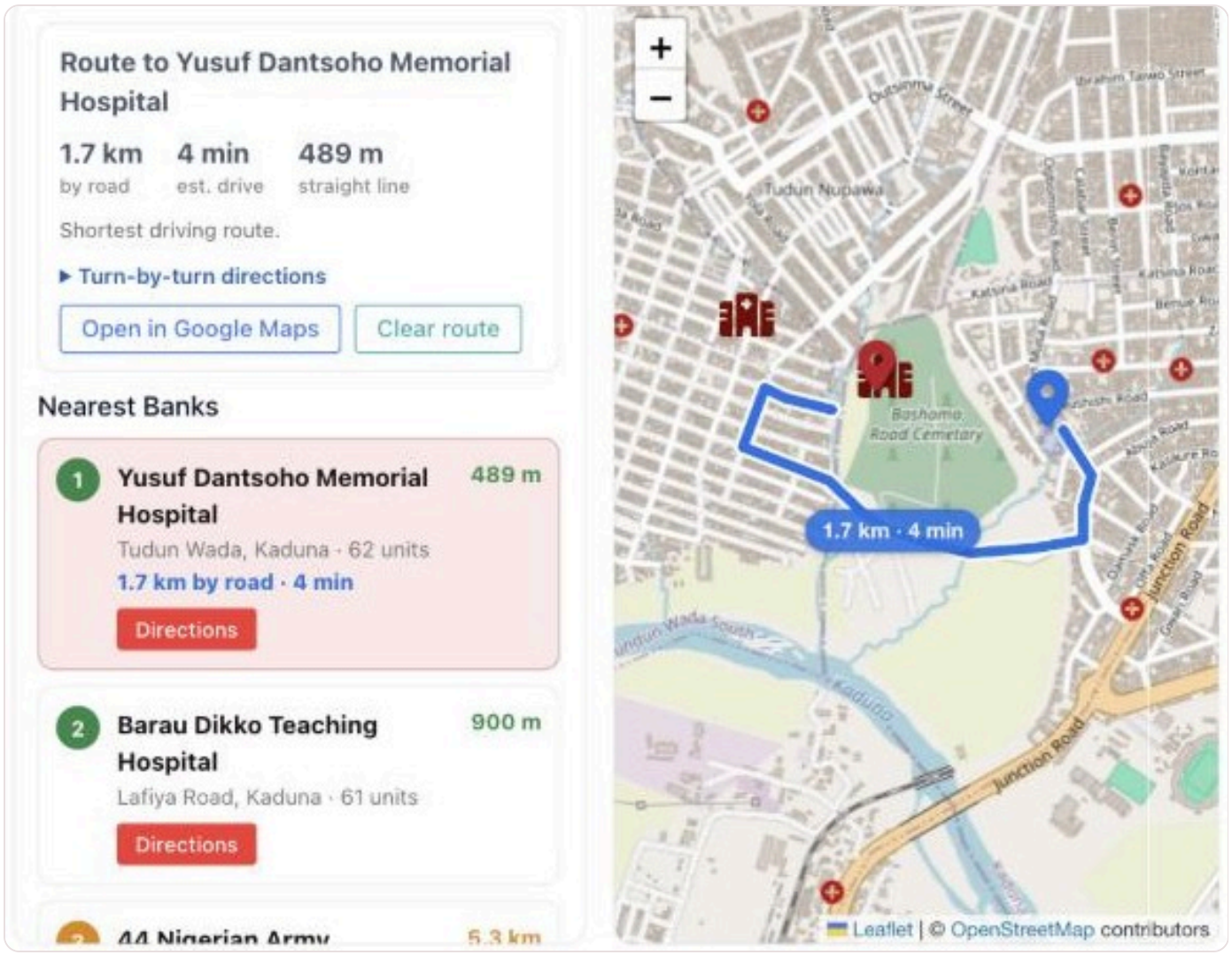
```
priority = severity       (critical 60 · severe 40 · moderate 20 · low 5)
         + rarity × 2     (AB− rarest 10 … O+ most common 3)
         + time pressure  (hours left ≤3: 20 · ≤12: 12 · ≤48: 6 · else 2)
         → capped at 100   # hours left = deadline − time elapsed, refreshed on every Board load
```

A Critical, AB−, needed-within-2-hours request scores 60 + 20 + 20 = 100 and leads the board. Compatibility score: 0 if incompatible, else 100 (exact) / 90 (donor O−) / 80 (same ABO) / 70 (otherwise compatible).

06 / Blood Bank Locator

The interactive map (brief Fig 5.5)

The locator answers "which registered bank is nearest, and what is the shortest way there?" It is available to every role (Nearby Banks for members, Geo-Map for staff).



The blue pin is you; the red pin is the chosen bank. The shortest road route is drawn in blue with its distance and drive time on the line; the panel repeats it next to the straight-line distance.

1. **Drop your pin**

   "Use my location" (browser GPS, with an accuracy ring), "Use Kaduna city centre", or **click anywhere on the map**. A blue "you are here" pin appears; drag it to adjust.

2. **Banks are ranked**

   Each bank's straight-line distance (Haversine) sorts the list nearest-first, with a coloured badge (green ≤ 5 km, amber 5–15 km, red > 15 km).

3. **The shortest route is drawn**

   BloodLink routes to the nearest bank automatically. OSRM returns up to three road routes; the **shortest by distance** is drawn in solid blue with a white outline and labelled on the line with its **kilometres and drive time**. Others appear as grey dashed lines — click one to use it.

4. **Pick another bank**

   Click a bank in the list, its "Directions" button, or its marker. A bank you choose stays selected when you move your pin; otherwise moving the pin re-routes to the new nearest bank.

5 **Follow it**

The panel shows road distance, drive time, straight-line distance, turn-by-turn directions, and an "Open in Google Maps" link for phone navigation. If the routing service is down, a straight line and its distance are shown instead.

Built on free, open services — Leaflet + OpenStreetMap tiles, Nominatim geocoding, OSRM routing — so there is no API key or billing account.

07 / The end-to-end donation

Follow one unit of blood

1 **A member files a Critical O− request**

needed within a few hours → a `blood_requests` row (Pending) with a high priority score and an automatic `notices` alert.

2 **Members see the alert**

on their Emergency Alerts page; the bell in the top bar shows a dot when there are alerts from the last 48 hours.

3 **Officer opens the Emergency Board**

the request sits at the top, its countdown running → Match Donor.

4 **System proposes donors**

only O−, 56-day-eligible members, available first, the requester excluded — each with a match score and distance to the hospital.

5 **Officer confirms (1–2 units)**

`process_donation()` re-checks everything, writes the donation with the **donor's** blood group and the officer's bank, and resets the donor's clock. The request becomes Approved, or Fulfilled once the units donated reach the units needed. Directed blood goes straight to the patient, so bank stock is unchanged.

6 **Everyone sees the result**

The requester's history shows the status and the donation received; the donor sees it under My Donations with their next eligible date; reports and dashboards update.

08 / QA runbook

How to test the whole app

Set up once, then run each role's script. The seed has 5 real banks, members on the 56-day boundary, all eight blood groups, two unavailable members, and pre-scored requests.

SETUP

```
mysql -u root -proot -e "CREATE DATABASE IF NOT EXISTS donor_app"
mysql -u root -proot donor_app < db/schema.sql
mysql -u root -proot donor_app < db/seed.sql
# existing database from v5? run once instead:  mysql -u root -proot donor_app <
db/upgrade_2026_10_qa.sql
```

ROLE	PAGE	USERNAME	PASSWORD
Admin	`login.php`	`admin`	`admin`
Officer	`user-login.php`	`maryam@bloodlink.test`	`Officer@123`
Member	`user-login.php`	`zainab@bloodlink.test` or `BL-KD-001`	`Member@123`

A. AUTHENTICATION & ACCESS

- ✓ All three logins land on the right dashboard immediately; wrong passwords are rejected; a member can log in with email or Member ID (any letter case).
- ✓ Logged out, a protected page redirects to the right login (admin pages → `login.php`); a logged-in user opening another role's page lands back on their own dashboard.
- ✓ A logged-in user opening a login or register page goes to their dashboard. Logout works from the topbar menu and the sidebar.

B. REGISTRATION

- ✓ A valid registration succeeds and shows the Member ID; the new member can log in.
- ✓ Each is refused with a clear message, and nothing is saved: duplicate email (any case), taken Member ID, blood group outside the eight, blank name, bad phone, password under 8 characters or without a number, mismatched confirmation.
- ✓ After an error, the typed values (except passwords) are still in the form.

C. ADMINISTRATOR

- ✓ **Blood Banks:** add (geocoded; 8 stock rows), edit, delete; a duplicate name shows a message, not an error page.
- ✓ **Members:** add, edit (password reset optional; untick "Available to donate"), delete; a member with donations is kept with an explanation.
- ✓ **Officers:** add one (bank required); a duplicate email is refused; an officer who has confirmed donations can't be deleted.
- ✓ **Blood Stock:** drop A+ below threshold → alert appears; −1 is refused; restock → alert resolves.

- ✓ **Charts & export:** dashboard and Reports charts render; the donations CSV opens as a clean spreadsheet.
- ✓ **Emergency Alerts:** post one containing "&" — it displays as "&" once; delete it.

D. MEMBER → EMERGENCY BOARD → OFFICER (THE SPINE)

1. **Member files O−, Critical Emergency, needed within 3 hours**
   - ✓ Pending in Request History; a matching alert exists; high priority score; invalid submissions refused.
2. **Officer opens the Emergency Board**
   - ✓ The new request ranks at or near the top; its countdown ticks.
3. **Officer matches**
   - ✓ Only compatible members; the requester isn't listed; unavailable members show a disabled "Unavailable" button; the day-30 member is excluded and the day-56 member included.
4. **Officer confirms**
   - ✓ The donation is recorded under the **donor's** group; units are capped at 1–2 and at what's still needed; a partly covered request shows Approved with "N still needed", and Fulfilled once covered; a Fulfilled request accepts no more donations.
5. **Walk-in donation**
   - ✓ Record Donation by email or Member ID → that bank's stock rises by the units; the member sees it under My Donations; a second walk-in inside 56 days is refused.

E. MEMBER SELF-SERVICE

- ✓ Availability: change status, schedule and contact preference → saved; refreshing the page doesn't resubmit.
- ✓ My Profile: change phone and address → saved and re-located on the map; an email already used by another member is refused.
- ✓ Change password: a wrong current password is refused; the new password works at the next login.
- ✓ My Donations lists matched and walk-in donations with totals and the next eligible date.

F. BLOOD BANK LOCATOR

- ✓ Click the map → a blue pin drops there, banks re-rank, and a blue route to the nearest bank appears, labelled with km and minutes.

- ✓ Drag the pin → the route redraws to the new nearest bank; after choosing a bank yourself, dragging keeps that bank.
- ✓ Where alternatives exist, the blue line is the shortest ("Shortest of N routes"); clicking a grey route switches to it and shows how much longer it is.
- ✓ The panel shows road km, drive time, straight-line km, turn-by-turn steps and an "Open in Google Maps" link.
- ✓ On a phone the map sits above the bank list with no sideways scrolling.

G. CROSS-CUTTING

- ✓ Every page loads for every role with no PHP warnings or console errors.
- ✓ Phone width collapses the sidebar to a menu button; the top bar's logo, bell and avatar don't overlap.
- ✓ A tampered or expired form token is rejected (CSRF); deleting an alert only works through its Delete button.

KNOWN LIMITATIONS (EXPECTED)

- **Nominatim** allows about one lookup per second: saving many addresses quickly may leave some without map coordinates until re-saved.
- The public **OSRM** demo server is fine for a demo, not production traffic; self-host it for real deployment.
- Browsers only share GPS location on **HTTPS** or `localhost`; elsewhere use the Kaduna button or click the map.
- Admins don't confirm donations by design — the officer is the clinical actor, and the donation's bank is the officer's bank.